

Danny Rollo

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EDUCATION

Northeastern University
M.S. in Computer Science
B.S. in Computer Science & Mathematics

Boston, MA
Sept 2024 – Expected May 2026
Sept 2021 – May 2025

PUBLICATIONS

Rollo, D. “Lightweight CNN Encoders for Nucleus Segmentation.” *MICCAI 2026*. **Accepted — poster.**

TECHNICAL SKILLS

Languages: Python, Go, TypeScript/JavaScript, SQL
Deep Learning: PyTorch, timm, CNNs, U-Net, Vision Transformers (Swin), Transformers & attention
Machine Learning: scikit-learn, InterpretML (EBM), gradient boosting, PCA, SVM, collaborative filtering
LLM & Agentic Systems: Claude and OpenAI APIs, MCP servers, multi-agent orchestration, RAG, SentenceTransformers, vector search
Infrastructure: Docker, AWS (S3), FastAPI, MongoDB Atlas, SLURM / HPC clusters

SELECTED PROJECTS

Lightweight CNN Encoders for Nucleus Segmentation PyTorch, timm, PanNuke, MoNuSeg
• Designed and ran a controlled encoder ablation isolating encoder choice in U-Net segmentation, holding decoder, augmentation, and optimization schedule fixed across VGG, ResNet, and Swin-T backbones.
• Established that a 34M-parameter VGG U-Net (Dice 0.851 on PanNuke; 0.796 on MoNuSeg) matches a purpose-built medical transformer and is Pareto-dominant on inference cost.

Two-Tower Recommendation with Cold Start Ablation PyTorch, BPR loss, Yelp Open Dataset
• Found context features alone (day-of-week, distance, restaurant temporal profiles) outperform the full model on cold-start users by 4.2 NDCG@10 points, while content features remain essential for cold restaurants — an asymmetry in feature importance across cold-start regimes.
• Designed a leave-one-out onboarding evaluation that prevents information leakage when a cold user’s only signal comes from held-out reviews.

RailMax: Amtrak Service Optimizer InterpretML, GTFS, Census ACS, Folium
• Modeled the US Amtrak network as a directed weighted graph and engineered node-level features including destination-population-weighted reachability and eigenvector centrality.
• Trained an Explainable Boosting Machine ($R^2 = 0.76$, RMSE 0.84 log scale) under 5-fold stratified group cross-validation with k-means geographic clustering to prevent spatial leakage.

EXPERIENCE

Teradata — AI/ML Product Management Analytics Intern June 2025 – May 2026
• Authored the business case and led the design of a context-aware AI Notebook Assistant for VantageCloud AI Studio, enabling intelligent code generation, contextual suggestions, and integration with company libraries and documentation.
• Built frontend prototypes and interaction specs for agentic notebook assistant features, translating LLM capability constraints into user-facing product decisions.

PlateMate — Co-Founder & Chief Technology Officer May 2025 – Present
• Lead ML architecture and engineering for a food recommendation platform: hybrid item-based collaborative filtering plus semantic vector search over restaurant reviews.
• Built a RAG retrieval pipeline with 768-dimensional `a11-mpnet-base-v2` embeddings on MongoDB Atlas Vector Search, parsing natural-language dish queries into semantic and structured geo-aware filters.

LineVision Inc. — Software Engineer Co-op Jan 2024 – June 2024
• Full-stack owner of an internal web application; defined features with stakeholders, expanded APIs and database schema, maintained the application end to end.
• Re-architected a data pipeline for staff scientists, achieving a 1000× speed improvement.

Generate Product Development — Project Lead, Technical Lead, Software Engineer Jan 2023 – May 2025
• Led 14 engineers and designers to deliver a production mobile application; defined requirements with external stakeholders.

Khoury College of Computer Sciences — Lead Teaching Assistant 2022 – 2025
Fundamentals of Computer Science I & II · Led weekly two-hour laboratory sections of 40+ students across six semesters, held 1-on-1 office hours, and graded assignments.

HONORS & AWARDS

Rookie of the Year, HackBeanpot · Students of the Year Runner-Up, LLS · Eagle Scout, Scouts BSA